Database Management System

BCA-102T

What is Database?

- The Database is a collection of inter-related data which is used to retrieve, insert and delete the efficiently. It is also used to organize the data in the form of a table, schema, view, and report etc.
- C- reate
- R-Read
- U- Update
- D- Delete

Basic Concept

- **Software:-** is a set of instructions, programs, or data that runs on a device to perform specific tasks.

- **Hardware:-** Collection of mechanical, electrical & electronic parts.

- **System software :-**is a type of computer program that manages a computer's hardware and application programs
- .
- **Application software:-** Developed as per the requirements. Ex. Word, Browser & Apps etc**.**

- **Data:-** Collection facts, figure, numbers etc.

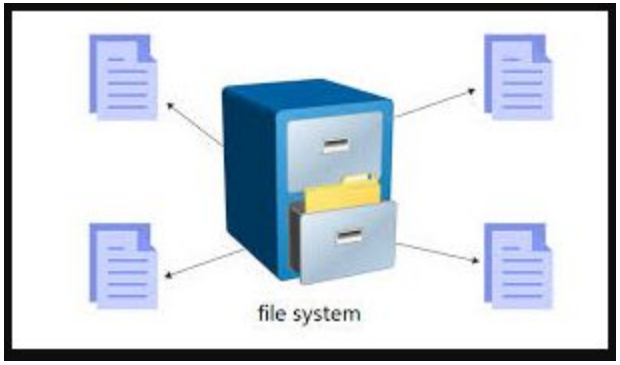
- **Database:-**Collections of interrelated data.

- **Manipulation:-** Operations perform in database.

About File System Vs DBMS

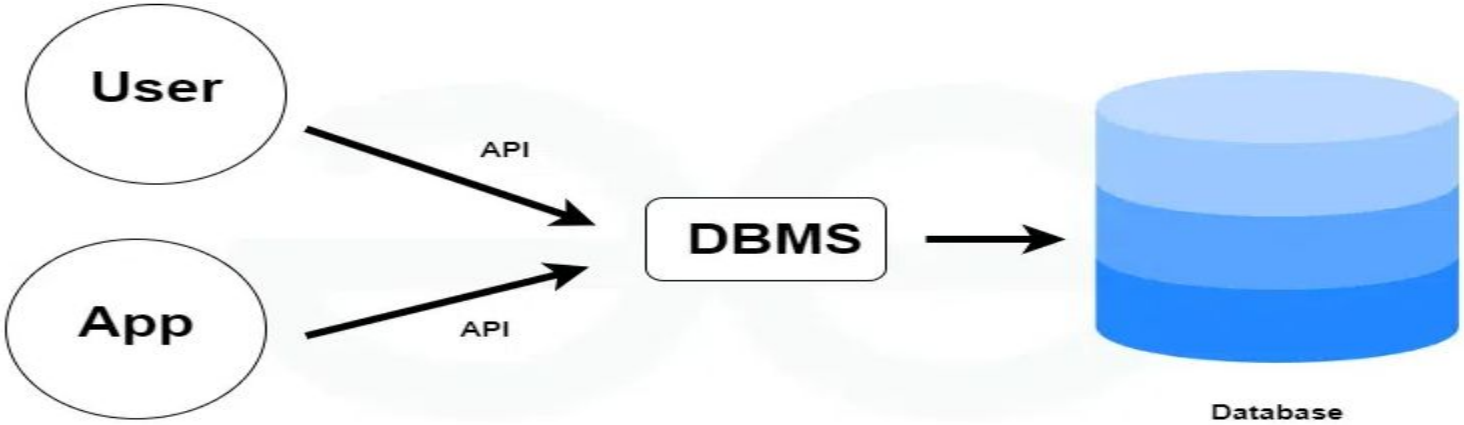
What is File System?

- A File management system is a DBMS that <u>allows access to single files or tables at a time.</u>
- In File System, Udta is directly stored in set of files.
- It contains <u>flat files that have no relation to other files</u>



What is DBMS?

- A Database Management System (DBMS) is a application software that allows users to efficiently define, create, maintain and share databases.
- Defining a databases involve specifying the data types, structures and constraints of the data to be stored in the database.



Real Life Applications of DBMS

- **Traditional Database:**
  - Banking applications
  - Hotels & Airline Reservations
  - Access online Libraries
  - Purchase different items online or supermarket regarding grocery items etc.
- **Real Time & Active Databases:**
  - Geographical information system for analyzed maps, weather data & satellite images.
  - Data warehouse & Online Analytical Processing System(OLAP) used to extract & analyzed useful business information from very large database to support decision making.

Online Analytical Processing(OLAP) refers to a set of software tools used for data analysis in order to make business decisions.

Advantages of DBMS over File System

1. Data Redundancy:
   - Redundancy is a concept of repetition of data i.e. each data may have more then a single copy.
   - The file system cannot control redundancy of data as each user defines and maintains the needed files for a specific application to run.
   - Whereas DBMS controls redundancy by maintaining a single repository of data that is defined once and is accessed by many users.

Advantages of DBMS over File System

2. **Data Sharing:**

- ✔ file system does not allow sharing of data or sharing is too complex.
- ✔ Whereas in DBMS, data can be shared easily due to a centralized system.
- ✔ Ex. Google Drive ,Google form etc.

3. **Data Integrity:-**

- ✔ There may be cases when some constraints need to be applied to the data before inserting it into the database.
- ✔ The file system does not provide any procedure to check these constraints automatically.
- ✔ Whereas DBMS maintains data integrity by enforcing user-defined constraints on data by itself.
- ✔ Ex: Age, Mobile No. etc.

Advantages of DBMS over File System

4. **Avoiding Inconsistency:**

- ✔ If any user changes the data in its files, then the changes are not reflected in all files. This creates inconsistency in the data.
- ✔ But in the DBMS, there is only one repository of data, and multiple users can use it. If any user changes the data, then it is reflected to each user as they are using the same repository.
- ✔ Ex:- A customer's address is updated in the Marketing team's system but not in the Sales team's system.

Advantages of DBMS over File System

4. **Enforcement of security:**

- ✔ Security enforced by DBA.
- ✔ DBA sets the security contraints to users for the common operations like adding, deleting, retrieving & updating of records.
- ✔ DBA assign different privileges to the users.

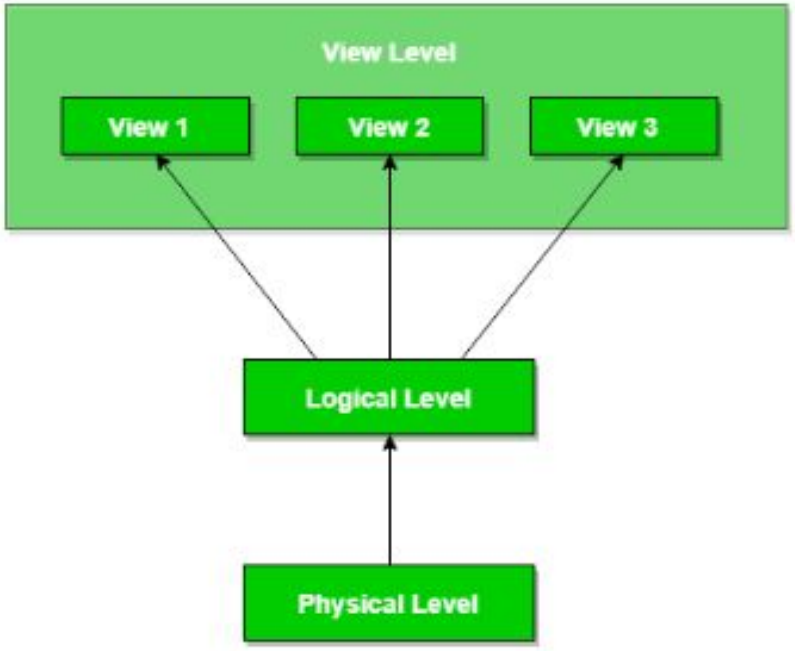
Data Abstraction & Data Independence

- Defination:
  - **Data abstraction** refers to the <u>process of hiding irrelevant details from the users.</u>
  - Example:
    - A switchboard provides a simple way to turn electrical appliances on and off, while hiding the details of the circuitry and current flow.

Data Independence : means <u>users & data should not directly interact with each other</u>.

- The <u>user should me at a different level and the data should be present at some other level.</u>

- There are mainly <u>three levels of data abstraction</u> and we divide it into three levels in order to achieve **Data Independence**.
- Physical or Internal Level
- Logical or Conceptual Level
- View or External Level



Data Abstraction

Level 1: **Physical or Internal Level**

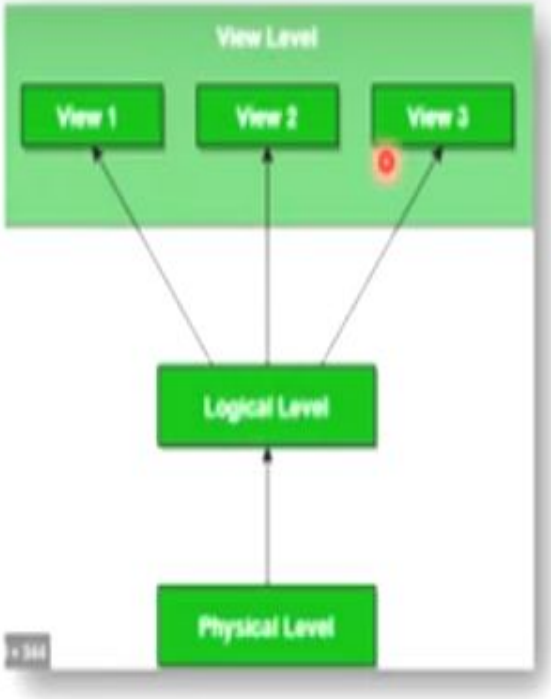
- This is the lowest level of data abstraction.
- It defines how the data is actually stored in memory
- It tells the actual location of the data that is being stored by the user.
- The database Administrator (DBA) is responsible for deciding how data is stored , which data should be kept at which particular disk drive, how the data has to arranged etc.
- They decide if the data has to be centeralized or distributed.
- It totally depends on the DBA how he/she manages the database at physical level.

Level 2: **Logical or Conceptual Level**

- The information is stored in the database in the form of tables.
- This level defines <u>what data are stored in database & What relationship exist among these data.</u>
- Logical levels decide structure of entire database.
- DBA use the <u>logical level for abstraction purpose</u>.

- Let us take an example
- A table named 'class' may have different attributes like student_name, Roll_no, student ID, and Marks.
- Though the data is stored in the Database but the structures of table like student table, teacher table, book table etc. are defined here in the conceptual level.
- Also how the tables are related to each other are defined here.

Level 3: View Level or External Schema

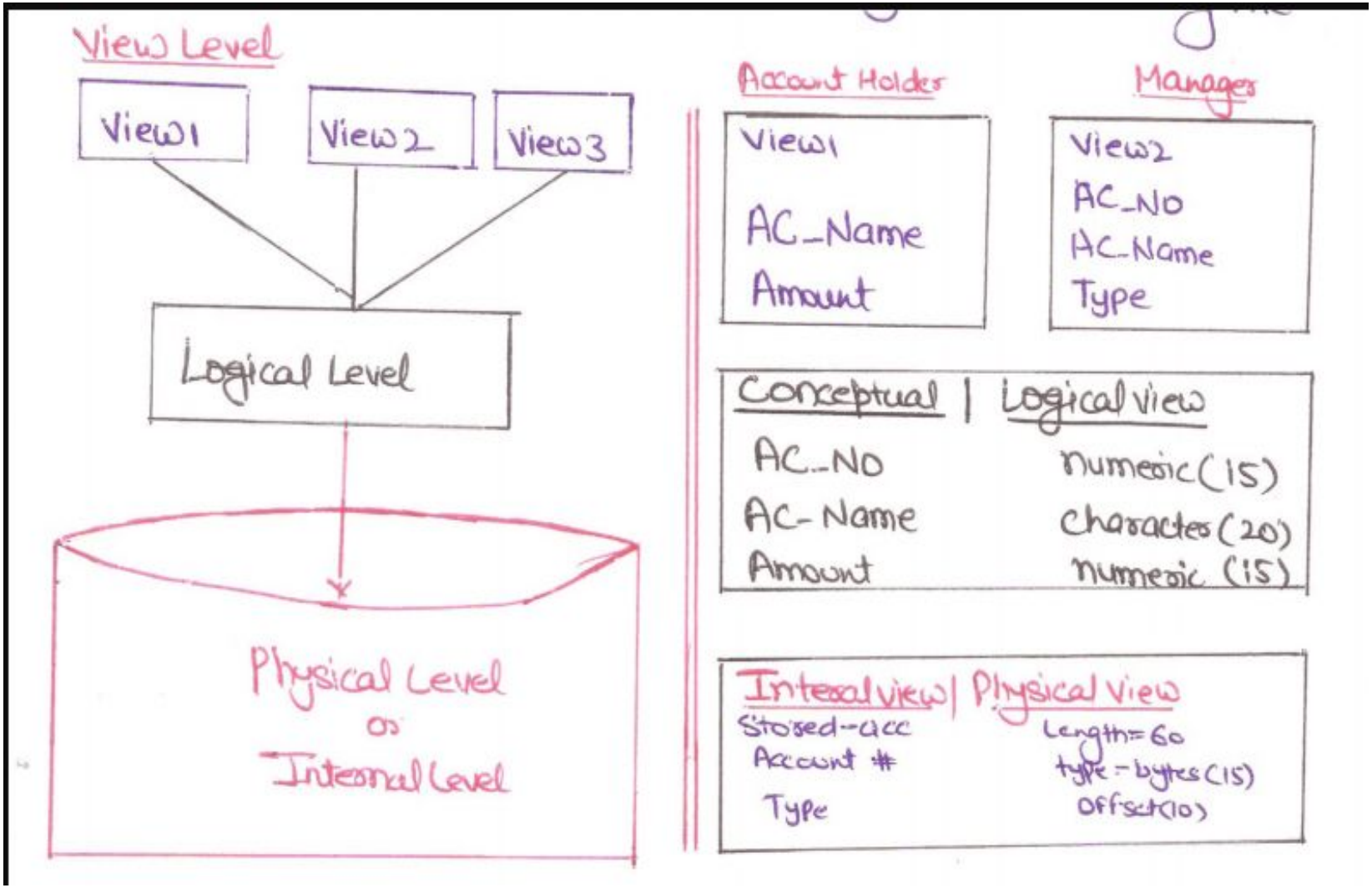


- This level tells the application about how the data should be shown to the user.
- Different views of same database can be created for user to interact with database for user friendly approach.

Example:

- If we have a login-id and password in a university system, then as a student, we can view our marks, attendance, fee structure, etc. But the faculty of the university will have a different view.
- He will have options like salary, edit marks of a student, enter attendance of the students, etc.
- So, both the student and the faculty have a different view.
- By doing so, the security of the system also increases.
- In this example, the student can't edit his marks but the faculty who is authorized to edit the marks can edit the student's marks.

Example of Data Abstraction in Database



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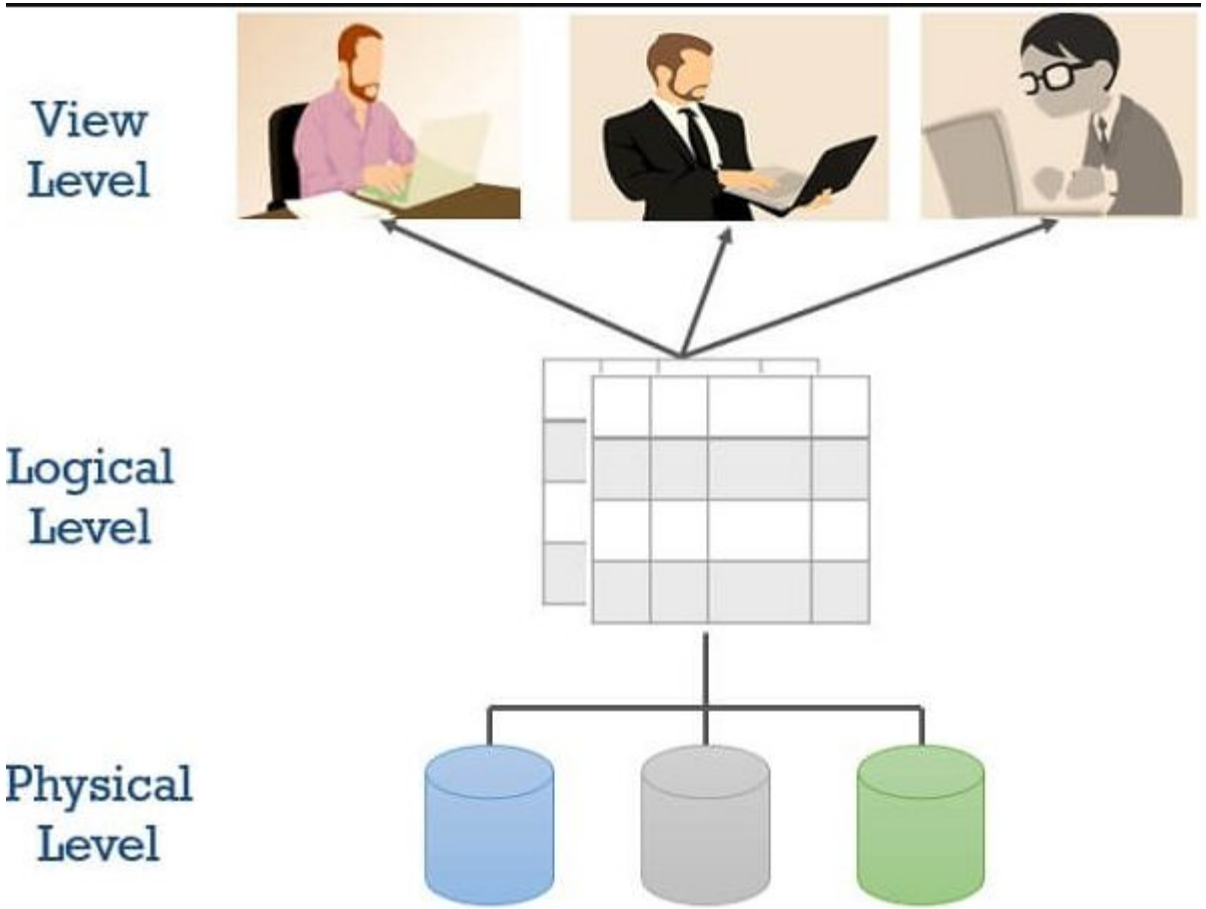
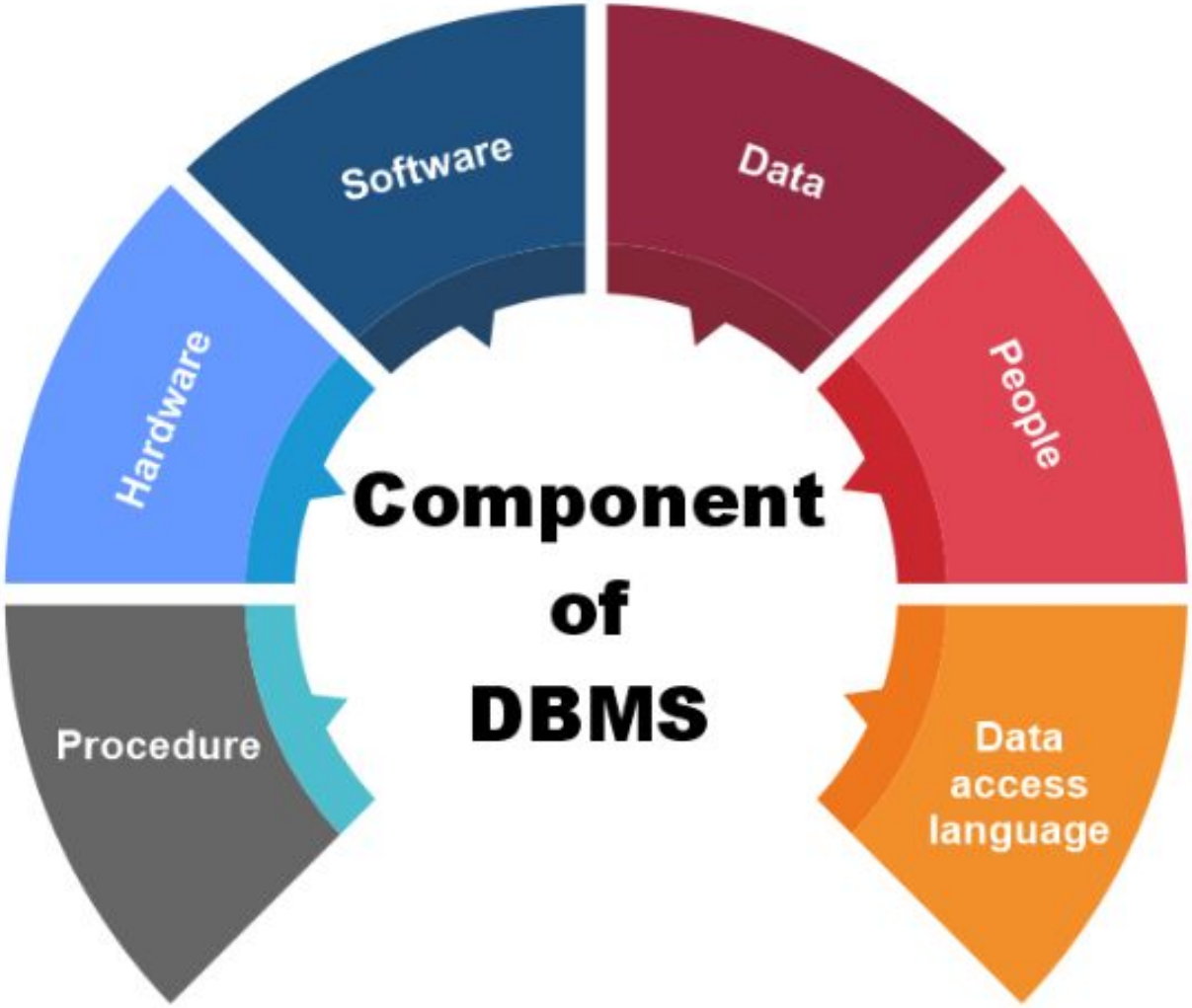


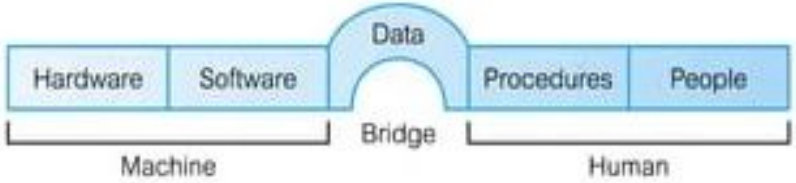
Components of DBMS



Components of DBMS

- **Hardware**: set of physical electronic devices that store the data, such as computers ,I/O devices and storage devices
- **Software**: This is a set of program that used to control & manages the overall database.
- **Data:** The main task of DBMS is to process the data.

Example: Name(Size, Character, Unique, Length)



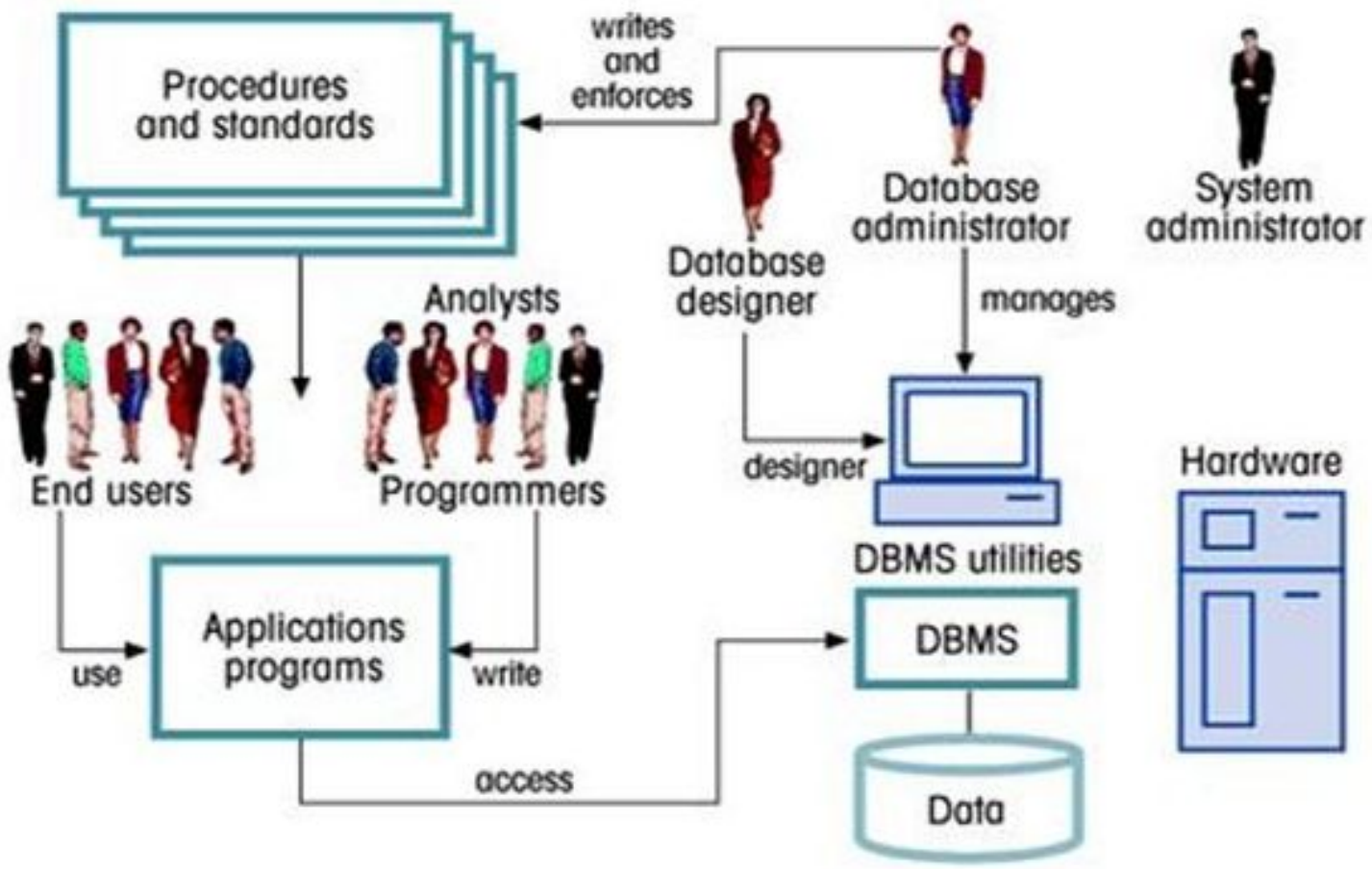
Components of DBMS

- **Procedures**: These are the instructions & rules that assist on how to use the DBMS and in designing and running the database, using documented procedures, to guide the users that operate and manage it.
- **People:**
  - The people who control and manage the databases and perform different types of operations on the database in the DBMS.
    - database administrator
    - software developer
    - End-user.

Components of DBMS

- **Database Access Language**
  - Database Access Language is a simple language that allows users to write commands to perform the desired operations on the data that is stored in the database.
  - Through utilizing the language, users can create new databases and tables, insert data and delete data.
  - Examples of database languages are SQL (structured query language), My Access, Oracle, etc. A database language is comprised of two languages.
    1. **Data Definition Language(DDL):**It is used to construct a database. DDL implements database schema at the physical, logical, and external levels.
    2. **Data Manipulation Language(DML):** It is used to access a database. The DML provides the statements to retrieve, modify, insert and delete the data from the database.

Components of DBMS



The Database System Environment

Characteristics of DBMS

- Self describing nature of a database system
- Insulation between programs and data, & data abstraction
- Support of Multiple views of the data.
- Sharing of data and multiuser transaction processing

Characteristics of DBMS

- **Real World Entity**
  - For example, if we want to create a student database, we need some entity. Any student stores their data.
  - In the Database, then, it should be the real-world entity. The most commonly used properties in the student database are name, age, gender, roll number, etc.

Characteristics of DBMS

- **Self-explaining nature**
  - In DBMS (Database Management System), the Database contains another database, and another database also contains metadata.
  - Here the term metadata means data about data.
  - So the self-explaining nature means the Database explains all the information automatically itself. This is because, in the Database, all the data are stored in a structured format.

Characteristics of DBMS

- **Insulation between Programs and Data, and Data Abstraction**
  - In a traditional file processing system, the structure of database knowledge files is embedded within the application programs, so any changes to the structure of a file may require changing all programs that access that file.
  - Against this, DBMS access programs don’t require such changes in most cases, so independence is achieved between them.

Characteristics of DBMS

- **Support for Multiple Views of the Data**
  - A database sometimes has many users, each of whom may require a special perspective or view of the database.
  - A view could also be a subset of the database.

- Integrity
  - Here the term integrity means the data should be correct and consistent in nature..

Characteristics of DBMS

- **Sharing of knowledge and Multi-user Transaction Processing**
  - A multi-user DBMS, as its name implies, must allow multiple users to access the database at an equivalent time or concurrently.
  - This is often essential if data for multiple applications is to be integrated and maintained during a single database such as the latest feature of WhatsApp integration with Facebook.

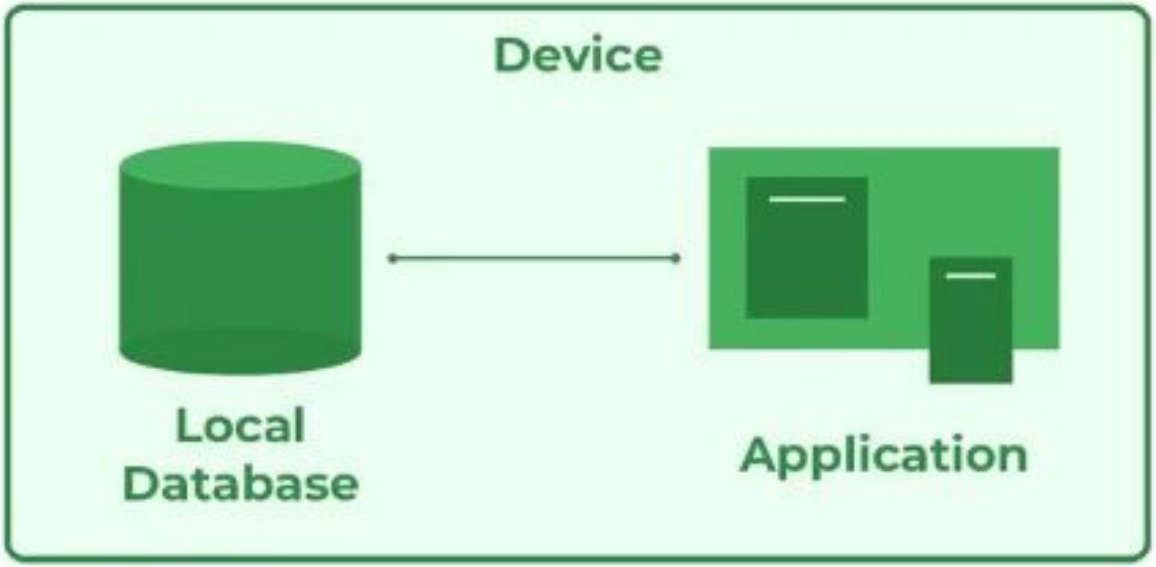
Types of DBMS Architecture

- 1-Tier Architecture
- 2-Tier Architecture
- 3-Tier Architecture

1-Tier Architecture

- In 1-Tier Architecture the database is directly available to the user
- the user can directly sit on the DBMS and use it that is, the client, server, and Database are all present on the same machine.
- This setup is simple and is often used in personal or standalone applications where the user interacts directly with the database.
- For Example: A Microsoft Excel spreadsheet is a great example of one-tier architecture.

1-Tier Architecture

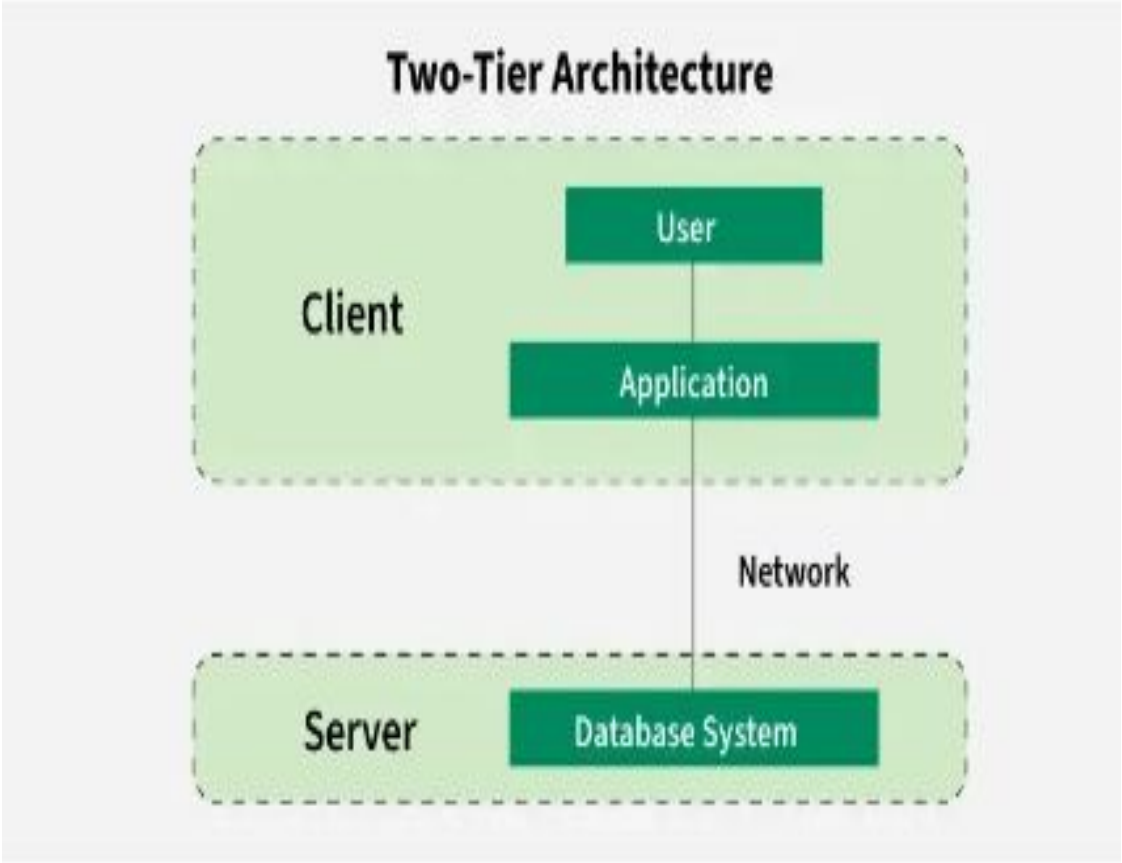


DBMS 1-Tier Architecture

2-Tier Architecture

- The 2-tier architecture is similar to a basic <u>client & server model</u>
- The application at the client end directly communicates with the database on the server side
- APIs like ODBC and JDBC are used for this interaction.
- For Example: A Library Management System used in schools or small organizations is a classic example of two-tier architecture.

- **Client Layer (Tier 1):** This is the user interface that library staff or users interact with. For example they might use a desktop application to search for books, issue them, or check due dates.
- **Database Layer (Tier 2):** The database server stores all the library records such as book details, user information, and transaction logs.
- The client layer sends a request (like searching for a book) to the database layer which processes it and sends back the result. This separation allows the client to focus on the user interface, while the server handles data storage and retrieval.



DBMS 2-Tier Architecture

Client Server Database & Its Applications

- Client server database consists of two logical components.
- One is “Client” and the other one is “Server”.
- Clients are those who send the request to perform a specific task to the server.
- Servers are normally receive the command sent by the clients, perform the task and send the appropriate result back to the client.

Client system vs Server system

- A client system is that who sends the request to the server system
- And the server system has to response the request as result.
- A client system is managed by users .
- where as the server system is managed by a computer expert.
- No need of a computer expert in client system.

Types of client server architecture

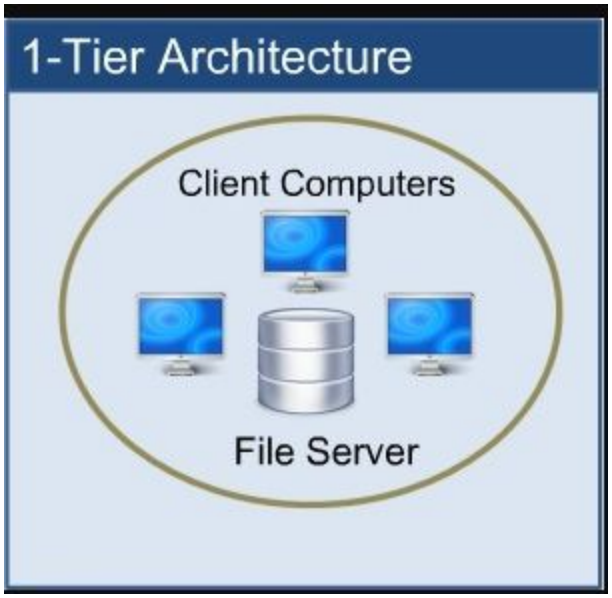
Normally, there are three types of client server architecture available in database management system.

These are listed below.

(a) Single tier client server computing model
(b) Two tier client server computing model
(c) Three tier client server computing model

Single tier client server architecture.

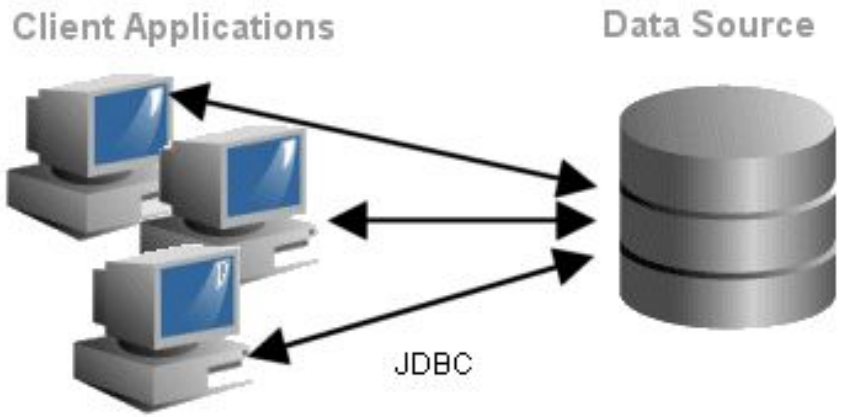
- It is the first type of client server computing model.
- In single tire system, the database is centralized, which means the DBMS software and the data in one location



Two-tier client server computing model

- It uses two computers and one client/server system.
- client is considered as data capture and validation tier and server was considered as data storage tire.

The most important limitation of the two-tier architecture is that it is not scalable, because each client requires its own database session.



Problems in two tier architecture

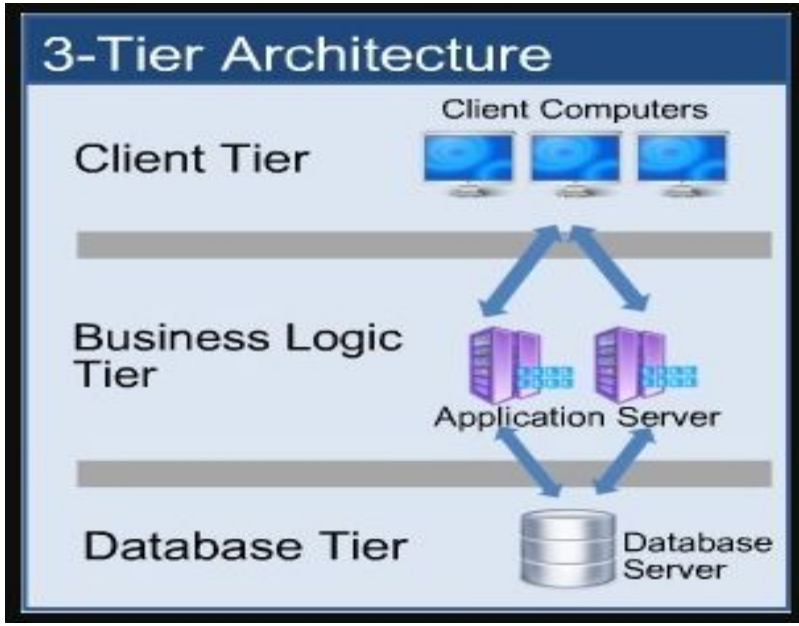
In 1990s, when applications were more complex and used with hundreds or thousands of end users, the two tier server model could not meet the need of the enterprise.

(a) An efficient client requiring considerable resources on client's computer to run effectively. This includes disk space, RAM, CPU.

(b) Client machines require administration which results overhead at the server end.

Three tier architecture.

- Three tier architecture has three layers.
- The first layer is the user interface which runs on client system.
- The second layer is called the application server .
- It is used for business logic and data processing.
- The third layer is known as database server. It is a database management system which stores the data as needed by the middle layer.

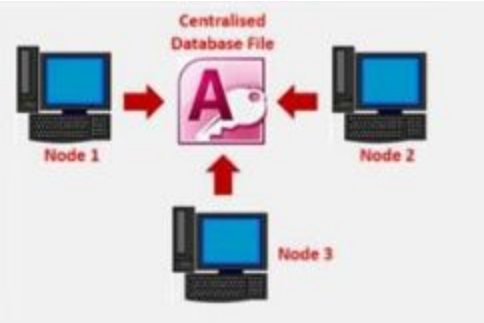


Data Models

- A data model in DBMS shows how data is stored, accessed, changed and connected.
- It also specifies the data items and the relationships between them.
- It defines the logical design and structure of a database.
- The data model sets the rule, relationship, constraints, etc. to define how data is stored in the database.
- It is like creating a blueprint of your Database.

Difference between Centralized and Distributed database

- In a **centralized database** system, all data is stored in a single location. This could be a single computer or server.
- **Advantages**:
  - **Simplicity**: It's easier to manage data as it's all in one place.
  - **Consistency**: Since there's only one copy of the data, it's always consistent.
- **Disadvantages**:
  - **Single Point of Failure**: If the central server fails, the entire system goes down.
  - **Scalability**: Bottle necking for multiple user s accessing the same file- slowing down productivity.



- In a **distributed database** system, data is spread across multiple locations, which could be on different servers or in different geographical locations.

•**Advantages**:

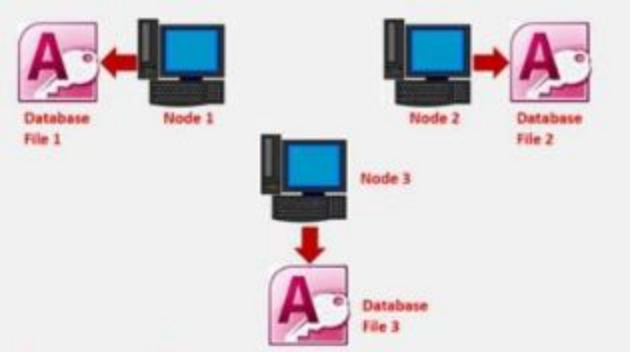
-Reliability: If one server fails, the system can still function using data from other servers.

-Scalability: It's easier to add more servers as the amount of data increases.

•**Disadvantages**:

✔**Complexity**: Managing data across multiple locations can be complex.

✔**Consistency**: Ensuring all copies of the data are consistent can be challenging.



Here's a comparison table for a quick overview:

	Centralized Database	Distributed Database
Data Location	Single location	Multiple locations
Simplicity	High	Low
Consistency	High	Can be challenging
Single Point of Failure	Yes	No
Scalability	Limited	High